

Arshia Fazeli

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Education

University of Waterloo

Sept 2021 – Sept 2026

PhD in Chemical Engineering

- GPA : 92.67/100
- **Coursework:** Theory and Application of Transport Phenomena, Applied Engineering Mathematics, Chemical Reactor Analysis, Computational Fluid Dynamics for Engineering Design, Multiphase Flows, Computational Fluid Dynamics, Consulting

University College London

Sept 2018 – June 2021

BEng in Chemical Engineering

- First Class Honour
- **Coursework:** Computational Modelling and Analysis, Reaction Engineering, Process Dynamics and Control, Particulate Systems and Separation Processes, Programming, Manufacture of Regenerative Medicine Products

Experience

Computational Multiphysics Researcher

Sept 2021 – Now

University of Waterloo

- Developed and validated a well-posed two-fluid model formulation with improved physical accuracy for simulating dispersed multiphase flow systems.
- Developed discontinuous Galerkin (DG) method solvers for various multiphysics problems, including two-fluid models, incompressible Navier–Stokes equations, and Cahn–Hilliard equation.
- Performed numerical verification of the developed solvers via the method of manufactured solution, assessing convergence rates and discretization error to confirm implementation accuracy.

Teacher Assistant

Sept 2021 – Now

University of Waterloo

- Teaching assistant for courses in transport phenomena, numerical methods for ODEs/PDEs, and computational methods.
- Developed and deployed a Python-based automated grading system for a programming course, using unittest, custom assertions, and regex parsing to evaluate submissions and deliver personalized feedback at scale.

Research Associate

Aug 2020 – Dec 2020

University College London

- Conducting molecular dynamics simulations using LAMMPS and PLUMED to investigate nucleation pathways in colloidal suspensions.

Projects

Discontinuous Galerkin based solvers for multiphysics problems

- Developed solvers based on the discontinuous Galerkin finite element method for a range of multiphysics problems, including the Euler–Euler model, mixture model, Cahn–Hilliard equation, and incompressible Navier–Stokes model.
- Numerical and code verification using the method of manufactured solutions.
- Tools Used: Python, NGSolve, NETGEN, PETSc, GMSH

Steady Incompressible Navier-Stokes Solvers with Scalar Transport and Reaction

SCGS [🔗](#), SIMPLE [🔗](#)

- Finite-volume method solvers for the incompressible Navier-Stokes equations, implemented using various algorithms for velocity-pressure coupling including SIMPLE and SCGS algorithms. Leveraged Numba JIT compilation to achieve high-performance execution and significant runtime speed-ups.

- SIMPLE-based solver extended to include scalar transport and reaction source terms.
- Implementation of multiple convection and diffusion flux schemes.
- Tools Used: Python, NumPy, Numba, Matplotlib, SymPy

Euler Equations Solver Using Flux Vector Splitting

[EULER-FVS](#) 

- Implemented a finite-volume method solver for the one-dimensional compressible Euler equations using various flux vector splitting schemes including Steger-Warming and Van Leer.
- Tools Used: Python, NumPy, SymPy, Matplotlib

CFD Simulation of a Side-Dump Combustor with Variable Inlet Angles

- Performed CFD simulations of a turbulent airflow in a side-dump combustor.
- Validated simulation results against experimental velocity and turbulence data.
- Investigated the impact of side-inlet angle on fuel-air mixing.
- Conducted mesh refinement and error analysis to quantify discretization and modelling errors.
- Tools Used: Ansys CFX

Publication (*Under Review*)

Laminar dispersion force effects on two-fluid modelling and simulation of bubble column hydrodynamics

[preprint](#) 

Arshia Fazeli, Sander Rhebergen, Nasser Mohieddin Abukhdeir

Volunteering

VP Academic at University of Waterloo Chemical Engineering Graduate Student Association (CEGSA)	Jan 2022 - Jan 2023
Welfare Officer at UCL Technology Society	Aug 2020 - July 2021
Outreach Volunteer at Engineers Without Borders	Oct 2019 - Apr 2020
Engineer at UCL Project Impactive	Oct 2019 - Jan 2020

Awards

CSCHE 2024 Best Poster Award	certificate  , Nov 2024
Garry L. Rempel Memorial Graduate Scholarship	May 2022
Best Final Design Project Team Award	June 2021

Technologies

Languages: Python, MATLAB, C++

Tools: Python, NumPy, Numba, NGSolve, PETSc, OpenFOAM, Ansys CFX, ParaView, Git, LaTeX, Linux, unittest, pytest